PROJECT REPORT

Retail Sales Analytics Dashboard

An end-to-end Power BI project on sales, profit, product, customer and regional performance

Tools: Power BI · SQL · Excel

Data: Superstore retail orders, 9,994 order lines, January 2014 – December 2017

Dashboard: 4 interactive report pages with slicers and business insights

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1. Executive Summary

This project analyses four years of retail order data (2014–2017) in Power BI and presents the results in a four-page interactive dashboard. It is designed to help a business leader answer four questions quickly: how are sales and profit trending, which products and categories earn their keep, who are the most valuable customers, and which regions perform best.

Headline Metrics

₹1.94M Total Sales	**₹222.16K** Total Profit	**4,172** Total Orders	**11.48%** Profit Margin
17.57% Average Discount	**792** Total Customers	**₹463.82** Average Order Value	**2014–2017** Analysis Period

Key Takeaways

- **Healthy but uneven profitability.** The business earns an 11.48% margin on ₹1.94M of sales, but Furniture converts only about 2% of its ₹642K sales into profit, compared with 15–17% for Office Supplies and Technology.
- **Technology drives the business.** It leads both sales (₹734K) and profit (₹125K). Copiers alone contribute the highest sub-category profit (₹52K).
- **Tables are loss-making.** Tables generate about ₹197K in sales but lose around ₹17K, and Bookcases and Supplies are also slightly negative.
- **Strong seasonality.** Sales peak in November (₹292K), with December and September also strong; January and February are the weakest months.
- **West and East lead.** West (₹582K) and East (₹576K) together generate more than 59% of total sales, while Central earns the lowest profit despite mid-range sales.

2. Project Overview

Objective

Build a clean, decision-ready Power BI report on top of a properly modelled dataset, so that sales performance, profitability, product mix, customer value and regional performance can be explored from a single file.

Scope

- Data preparation, validation and cleaning using Excel, SQL and Power BI
- A star-schema data model with a fact table, four dimension tables and a dedicated measures table
- DAX measures for sales, profit, margin, discount, orders, customers and average order value
- A four-page dashboard with KPI cards, charts, a map, slicers and written business insights on every page

Intended Audience

Sales managers, regional heads and business leadership who need a fast, visual summary of performance and a clear view of where to act.

3. Dataset

The source is a retail orders dataset (Superstore) with one row per order line. The file is included in this repository as Retail_Sales_Analytics_Dashboard.csv.

Property	Detail
Size	9,994 rows × 21 columns
Period	3 January 2014 – 30 December 2017
Data quality	No missing values and no duplicate rows
Date format	M/D/YYYY (Order Date, Ship Date)
File encoding	Windows-1252 / Latin-1 (use encoding="latin-1" when reading with pandas)

Columns

Group	Columns	Description
Order	Row ID, Order ID, Order Date, Ship Date, Ship Mode	Order identity, dates and shipping method
Customer	Customer ID, Customer Name, Segment	Who bought; Consumer, Corporate or Home Office
Location	Country, State, City, Postal Code, Region	Delivery location; Region is East, West, Central or South
Product	Product ID, Product Name, Category, Sub-Category	Furniture, Office Supplies and Technology, with 17 sub-categories
Financials	Sales, Quantity, Discount, Profit	Revenue, units, discount rate and profit per order line (negative profit means a loss)

4. Measures

All KPIs on the dashboard are DAX measures stored in the All Measures table.

Measure	Definition	Dashboard value
Total Sales	Sum of Sales across the filtered order lines	₹1.94M
Total Profit	Sum of Profit across the filtered order lines	₹222.16K
Profit Margin %	Total Profit divided by Total Sales	11.48%
Average Discount %	Average discount rate applied	17.57%
Total Orders	Distinct count of Order ID	4,172
Total Customers	Distinct count of customers	792
Average Order Value	Total Sales divided by Total Orders	₹463.82

Values are shown with no slicer selected (Year, Category, Segment and Region set to All).

5. Data Model

The report is built on a star schema. A single fact table holds the order-line transactions, and four dimension tables describe customers, geography, products and dates. Each dimension connects to the fact table with a one-to-many relationship, and every measure lives in a separate table named All Measures.

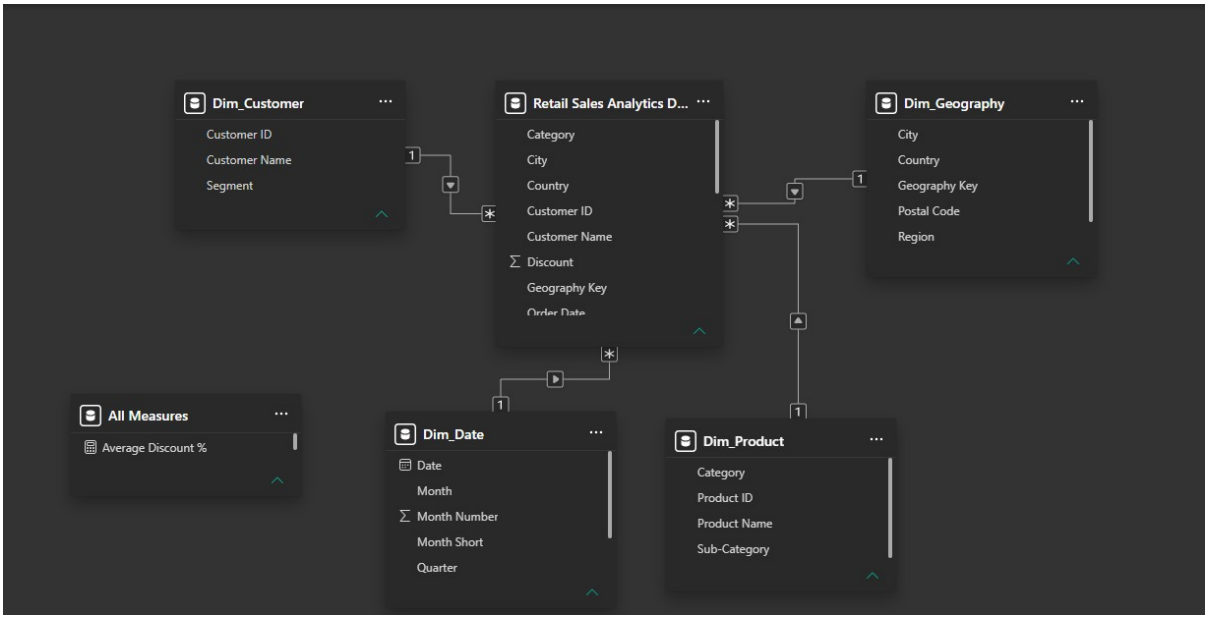


Figure 1: Power BI data model (star schema)

Table	Role	Key fields
Retail Sales Analytics Dashboard	Fact	Order ID, Order Date, Sales, Quantity, Discount, Profit and keys to each dimension
Dim_Customer	Dimension	Customer ID, Customer Name, Segment
Dim_Geography	Dimension	Geography Key, City, State, Region, Country, Postal Code
Dim_Product	Dimension	Product ID, Product Name, Category, Sub-Category
Dim_Date	Dimension	Date, Year, Month, Month Number, Month Short, Quarter
All Measures	Measure table	Total Sales, Total Profit, Profit Margin %, Average Discount %, Total Orders, Total Customers, Average Order Value

Benefits of this design: slicers filter every visual consistently, dimension attributes are stored once, and measures are easy to find, reuse and maintain.

6. Dashboard Walkthrough

The report has four pages. Each page has its own slicers and ends with a Key Business Insights panel that summarises the main message in plain language.

6.1 Executive Overview

Purpose: A one-page health check of the business.

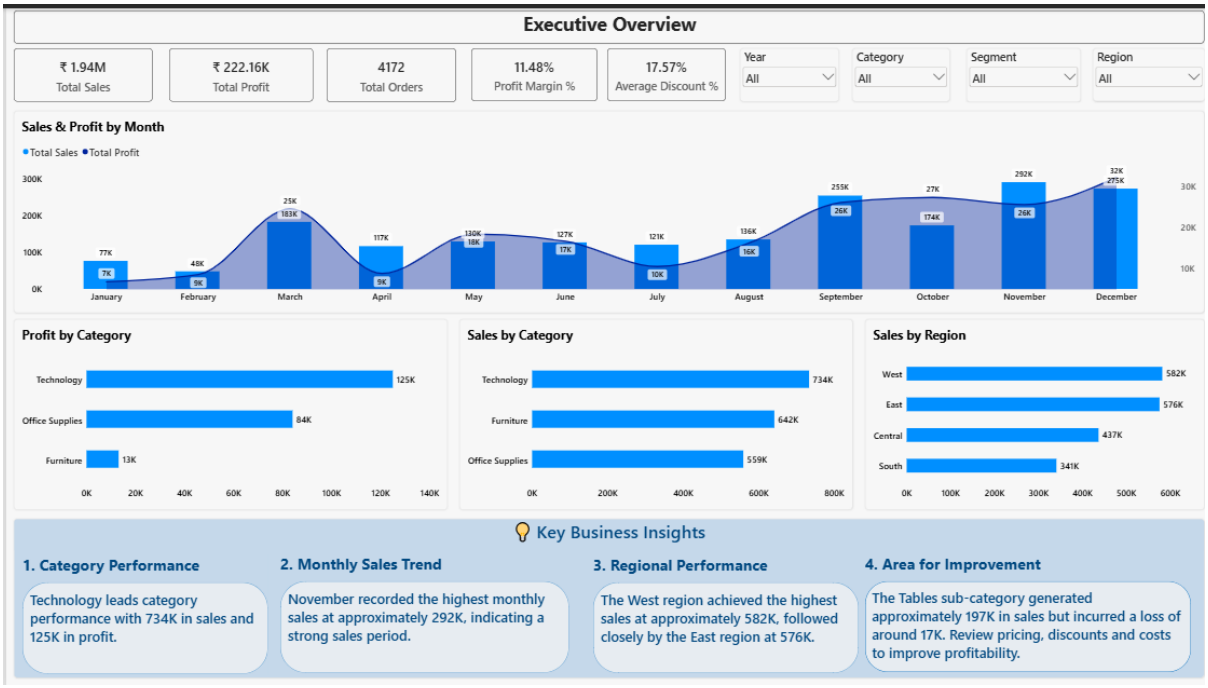


Figure 2: Executive Overview page

Visuals: five KPI cards, Sales & Profit by Month (combo chart), Profit by Category, Sales by Category, Sales by Region, and dropdown slicers for Year, Category, Segment and Region.

What the page shows

- Technology leads on both sales (₹734K) and profit (₹125K); Furniture is second on sales (₹642K) but last on profit (₹13K).
- Monthly sales peak in November at about ₹292K, followed by December (about ₹275K) and September (about ₹255K).
- West (₹582K) and East (₹576K) are close at the top, with Central (₹437K) and South (₹341K) behind.

6.2 Sales & Product Analysis

Purpose: Find the products and sub-categories that earn money, and those that lose it.



Figure 3: Sales & Product Analysis page

Visuals: Sales and Profit by Sub-Category, Sales and Profit by Category, Top 10 Products by Sales, and tile slicers for Year, Category and Segment.

What the page shows

- Phones (₹280K) and Chairs (₹267K) are the best-selling sub-categories.
- Copiers earn the highest profit (₹52K), followed by Phones (₹37K) and Accessories (₹32K).
- Three sub-categories lose money: Tables (about −₹17K), Bookcases (about −₹3K) and Supplies (about −₹1K).
- The Canon imageCLASS 2200 Advanced Copier is the top product at about ₹62K, far ahead of the next product (about ₹23K).

6.3 Customer Analysis

Purpose: Understand who the customers are and which ones create the most value.



Figure 4: Customer Analysis page

Visuals: KPI cards (Total Customers, Total Sales, Total Profit, Average Order Value), Sales by Customer Segment, Top 10 Customers by Sales, Customer-wise Profit, a donut chart of Customers by Segment, and slicers for Year, Segment and Region.

What the page shows

- The dashboard tracks 792 customers with an average order value of ₹463.82.
- Consumer is the largest segment: about ₹972K in sales and 409 customers (51.64%), ahead of Corporate (₹590K, 235 customers) and Home Office (₹373K, 148 customers).
- Sean Miller is the top customer by sales (about ₹24.5K), but does not appear among the top 10 customers by profit.
- Tamara Chand is the most profitable customer (about ₹8.6K), followed by Raymond Buch (₹7.0K) and Hunter Lopez (₹5.6K).

6.4 Regional Performance

Purpose: Compare regions, states and cities on sales and profit.

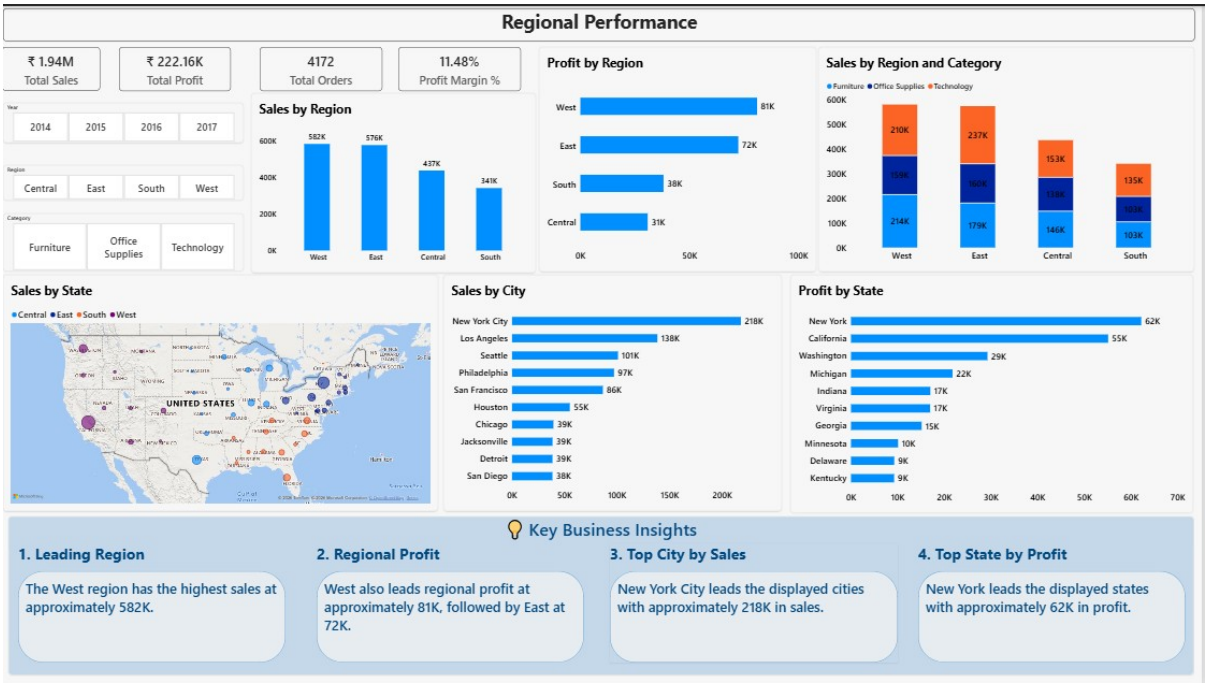


Figure 5: Regional Performance page

Visuals: KPI cards, Sales by Region, Profit by Region, Sales by Region and Category (stacked), a state-level map, Sales by City, Profit by State, and slicers for Year, Region and Category.

What the page shows

- West leads on both sales (₹582K) and profit (₹81K); East follows with ₹576K sales and ₹72K profit.
- New York City is the top city by sales (about ₹218K), ahead of Los Angeles (₹138K) and Seattle (₹101K).
- New York is the top state by profit (about ₹62K), followed by California (₹55K) and Washington (₹29K).
- Technology is the largest category in East (₹237K); West is the most evenly split between Furniture (₹214K) and Technology (₹210K).

7. Insights Summary

Area	Finding	Source page
Profitability	Overall margin is 11.48%. Furniture earns about 2% on ₹642K of sales, versus about 17% for Technology and 15% for Office Supplies.	Executive Overview
Seasonality	November is the strongest month (₹292K); January (₹77K) and February (₹48K) are the weakest.	Executive Overview
Products	Phones lead sales (₹280K); Copiers lead profit (₹52K). Tables lose about ₹17K on ₹197K of sales.	Sales & Product Analysis
Top product	Canon imageCLASS 2200 Advanced Copier sells about ₹62K, roughly 2.7 times the next product.	Sales & Product Analysis
Customers	Consumer drives about half of sales and customers. The top customer by sales is not among the top 10 by profit.	Customer Analysis
Regions	West and East account for more than 59% of sales. Central has the lowest profit (₹31K) despite ₹437K in sales.	Regional Performance
Locations	New York City is the top city by sales; New York is the top state by profit.	Regional Performance

Percentages and ratios in this table are calculated from the values shown on the dashboard.

Recommendations

1. **Fix Furniture profitability.** Review pricing, discounts and costs on Tables and Bookcases; Furniture earns about 2% margin against 15–17% elsewhere.
2. **Grow the high-margin lines.** Prioritise Copiers, Phones and Accessories, which together deliver the largest sub-category profits.
3. **Plan around seasonality.** Stock up and campaign ahead of September–December, and use targeted offers to lift the weak January–February period.
4. **Review the Central region.** Central converts about 7% of sales into profit, against about 14% in West; check discounting and cost drivers there.
5. **Look beyond sales when managing customers.** Review the profitability of high-sales accounts, and protect the most profitable customers such as Tamara Chand and Raymond Buch.
6. **Reduce dependence on one product.** One copier model contributes about ₹62K; keep availability high and build bundles with related supplies.

8. Conclusion & Next Steps

The dashboard turns raw order lines into a clear story: Technology and the West and East regions carry performance, Furniture (especially Tables) drags on profit, and sales are concentrated in the last quarter of the year. Every page is interactive, so users can slice by year, category, segment and region to explore these patterns themselves.

Possible Enhancements

- Add year-over-year growth and a discount-versus-profit view to explain the loss-making products
- Add drill-through pages from category to product and from region to city
- Publish the report to Power BI Service with scheduled refresh
- Add tooltips and bookmarks for a guided executive view

9. Tools Used

Tool	How it was used
Power BI	Data modeling (star schema), DAX measures, and a four-page report with KPI cards, combo, bar, column, donut and map visuals, slicers and cross-filtering
SQL	Querying and validating the retail order data and supporting the analysis
Excel	Reviewing and preparing the source CSV data

About the Author

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